Stockholm University

Department of Meteorology

Numerical Methods in Atmospheric Sciences and Oceanography (MO8007)

Lab 2 – The Shallow Water Equation

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1 Introduction

The shallow water equations are an important tool in oceanography and weather models to evaluate the behaviour of geophysical flows. This is a common type of equations since the horizontal extent of both the ocean and the atmosphere is much larger than their vertical extent, which allows for a simplification of the Navier-Stokes equations as shallow. Furthermore, these equations are much simpler than the full Navier-Stokes equations, which makes them ideal for testing and evaluating numerical schemes. In this report the shallow water equation will be solved using two different numerical grids, the unstaggered grid and the staggered grid. The theory presented in this report will be bases on Döös et al. [2022]. The result from this report will give insight into the world of more advanced numerical schemes, and maybe give insight into why staggered grids are more commonly used in weather models.

1.1 Theory

The shallow water equations are

$$\begin{cases} \dfrac{\partial u}{\partial t} + g\dfrac{\partial h}{\partial x} = 0 & \text{(1a)} \\ \dfrac{\partial h}{\partial t} + H\dfrac{\partial u}{\partial x} = 0. & \text{(1b)} \end{cases}$$

The analytical solution to the shallow water equations, assuming periodic boundary conditions and initial velocity of $u(x,0) = 0$, can be written as two opposing travelling waves

$$\begin{cases} h(x,t) = h_1(x - c\,t) + h_2(x + c\,t), \\ u(x,t) = u_1(x - c\,t) + u_2(x + c\,t), \end{cases}$$

where $c = \sqrt{gH}$ is the phase speed of the waves. The initial condition $h(x,t) = h_0(x)$ and $u(x,0) = 0$ can be used together with Equation 1 to find h_1, h_2, u_1 and u_2. From the initial condition, we can find that $h_1(x) + h_2(x) = h_0(x)$, and from the initial velocity we can find that $u_1(x) + u_2(x) = 0$. Using Equation 1 one can derive that $u_1(x) = \frac{c}{H}h_1(x)$ and $u_2(x) = -\frac{c}{H}h_2(x)$. This gives the following solution for the shallow water equations:

$$\begin{cases} h(x,t) = \dfrac{1}{2}h_0(x - c\,t) + \dfrac{1}{2}h_0(x + c\,t), & \text{(2a)} \\ u(x,t) = \dfrac{c}{2H}h_0(x - c\,t) - \dfrac{c}{2H}h_0(x + c\,t). & \text{(2b)} \end{cases}$$

1.2 The numerical models

1.2.1 Leapfrog scheme on an unstaggered grid

Using the leapfrog scheme for an unstaggered grid, the shallow water equations (Equation 1) can be discretized as:

$$\begin{cases} \dfrac{u_j^{n+1} - u_j^{n-1}}{2\Delta t} = -g\dfrac{h_{j+1}^n - h_{j-1}^n}{2\Delta x} & \text{(3a)} \\ \dfrac{h_j^{n+1} - h_j^{n-1}}{2\Delta t} = -H\dfrac{u_{j+1}^n - u_{j-1}^n}{2\Delta x}, & \text{(3b)} \end{cases}$$

where the grid was defined as $x_j = j\Delta x$ and $t^n = n\Delta t$. And the courant number μ is defined as $\mu = \frac{c\,\Delta t}{\Delta x}$. This gives the following step equations for u and h:

$$\begin{cases} u_j^{n+1} = u_j^{n-1} - \dfrac{g\Delta t}{\Delta x}\left(h_{j+1}^n - h_{j-1}^n\right) & \text{(4a)} \\ h_j^{n+1} = h_j^{n-1} - \dfrac{H\Delta t}{\Delta x}\left(u_{j+1}^n - u_{j-1}^n\right) & \text{(4b)} \end{cases}$$

One can note that this is stable if $\mu \leq 1$.

However, this scheme uses the fact that we have the values of u and h at time steps $n-1$ and n, which means that we need to initialize the model somehow. A common way of doing this is using a forward Euler step for the first time step, which gives:

$$u_j^1 = u_j^0 - \frac{g\Delta t}{2\Delta x}\left(h_{j+1}^0 - h_{j-1}^0\right) \tag{5a}$$

$$h_j^1 = h_j^0 - \frac{H\Delta t}{2\Delta x}\left(u_{j+1}^0 - u_{j-1}^0\right) \tag{5b}$$

1.2.2 Leapfrog scheme on an staggered grid

If one instead uses a leapfrog scheme with a staggered grid, the shallow water equations (Equation 1) can be discretized as:

$$\begin{cases} \dfrac{u_j^{n+1} - u_j^{n-1}}{2\Delta t} = -g\dfrac{h_{j+1}^n - h_j^n}{\Delta x} & \text{(6a)} \\ \dfrac{h_j^{n+1} - h_j^{n-1}}{2\Delta t} = -H\dfrac{u_{j+1}^n - u_j^n}{\Delta x}, & \text{(6b)} \end{cases}$$

where the grid now uses $x_j = j\Delta x$ for h_j and $x_{j+\frac{1}{2}} = (j+\frac{1}{2})\Delta x$ for u_j (if one uses the same j-coordinate for u as in h). This gives the following step equations for u and h:

$$\begin{cases} u_j^{n+1} = u_j^{n-1} - \dfrac{2g\Delta t}{\Delta x}\left(h_{j+1}^n - h_j^n\right) & \text{(7a)} \\ h_j^{n+1} = h_j^{n-1} - \dfrac{2H\Delta t}{\Delta x}\left(u_{j+1}^n - u_j^n\right) & \text{(7b)} \end{cases}$$

This scheme is stable if $\mu \leq 0.5$. However, it also needs to be initialized, and the same forward Euler step method can be used for this scheme.

1.2.3 Robert-Asselin filtering

There are different ways to get rid of this computational mode and thus get cleaner results. One common way is to use some kind of filter, such as the Robert-Asselin filter. The Robert-Asselin filter works by smoothing the result, since the computational mode adds background noise. This is done as

$$u_{\text{filtered}}^n = u^n + \gamma\left(u_{\text{filtered}}^{n-1} - 2u^n + u^{n+1}\right), \tag{8}$$

where γ is a filtering parameter. The Robert-Asselin filter has the drawback in that a it also dampens the result by a factor $[1 - 4\gamma\sin(\omega\Delta t/2)]$ due to the dispersion-like smoothing. Furthermore, this parameter also affects the stability criteria so that, for example, in the unstaggared Leapfrog scheme the stability condition becomes $\mu + \gamma \leq 1$. For the staggered Leapfrog scheme, the stability condition becomes $\mu + \gamma \leq 0.5$.

1.3 Error computation

The error between a numerical scheme and analytical solution can be quantified using the norm. For a vector $\mathbf{h}^n$, the ℓ_2-norm is defined as

$$\|\mathbf{h}^n\| = \sqrt{\frac{1}{N}\sum_{j=1}^{N}(h_j^n)^2}. \tag{9}$$

To obtain a dimensionless measure of the error, the relative error can be computed at time step n as

$$\text{error}^n = \frac{\|\mathbf{h}_{\text{numerical}}^n - \mathbf{h}_{\text{analytical}}^n\|}{\|\mathbf{h}_{\text{analytical}}^n\|}. \tag{10}$$

2 Methodology

The numerical solution of the shallow water equations was implemented using two different grids: an unstaggered grid and a staggered grid. The initial condition for the height field h was defined as a localized cosine wave perturbation given by

$$h_0(x) = \begin{cases} 0.5 + 0.5\cos\left(10\pi(x-0.5)\right), & 0.4 < x < 0.6, \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

And a velocity field u initialized as zero. Both numerical schemes were implemented using the leapfrog time-stepping method, for a total simulation time of $t = 1$ arbitrary time units and a spatial domain of $x \in [0, 1]$ arbitrary length units, with periodic boundary conditions. Both models used a phase speed of $c = 1$, mean fluid depth of $H = 1$ and a gravitational acceleration of $g = 1$.

The unstaggered grid scheme was implemented with a spatial resolution of $\Delta x = 0.025$ and a a Courant number of $\mu = 0.9$. The staggered grid scheme was implemented with three different spatial resolutions: $\Delta x = 0.05$, $\Delta x = 0.025$ and $\Delta x = 0.0125$, all with a Courant number of $\mu = 0.45$. A robert-Asselin filter with a filtering parameter of $\gamma = 0.005$ was applied to the staggered grid scheme, while a filter with $\gamma = 0.01$ was applied to the unstaggered grid scheme in order to evaluate the effect of the filter on these numerical solutions.

The relative error of the different schemes were computed at each time step by comparing the numerical solution to the analytical solution of the shallow water equations, which is given by

3 Results

The result from the unstaggered grid scheme is shown in Figure 1. The solutions shows to propagating waves in both directions, just like the theoretical solution (see Equation 2). However, the solution is affected by a numerical error, which grows in time. This can be seen by the edge of the pulse showcasing an oscillatory pattern. The solution also shows a small amount of numerical dispersion, since the wave become blurry with time.

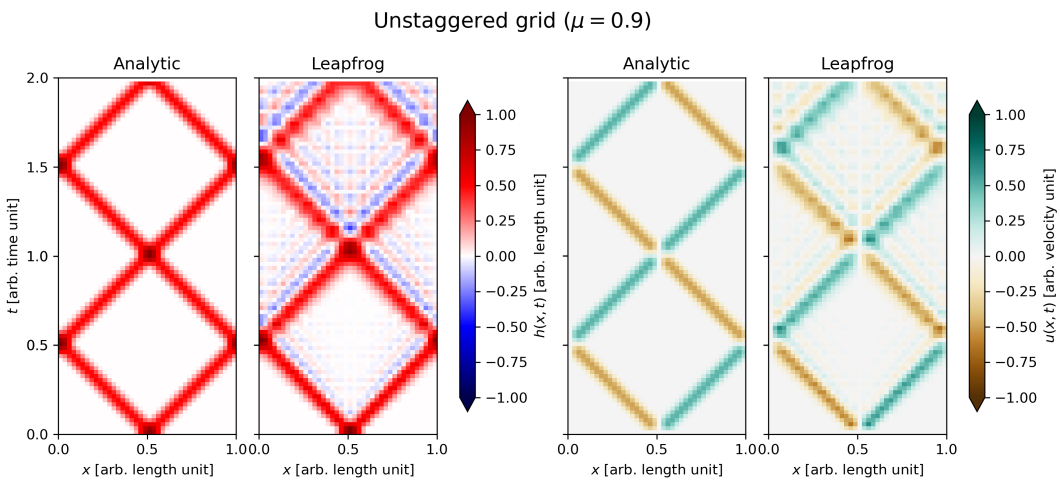

Figure 1: Hovmöller diagram of the numerical solution using the unstaggered grid. The propagation of the wave $h(x, t)$ and its velocity $u(x, t)$ is shown over time.

In Figure 2 the solution from the staggered grid scheme is shown. The solution shows a much better agreement with the theoretical solution, for the same grid spacing. One can note that this solution uses a smaller Courant number than the unstaggered grid solution since the staggered grid scheme needs a

half as large Courant number. The solution with the coarsest spatial resolution ($\Delta x = 0.05$) shows a significant amount of numerical error, while the solution with the finest spatial resolution ($\Delta x = 0.0125$) shows a much cleaner solution, with no noticeable numerical errors.

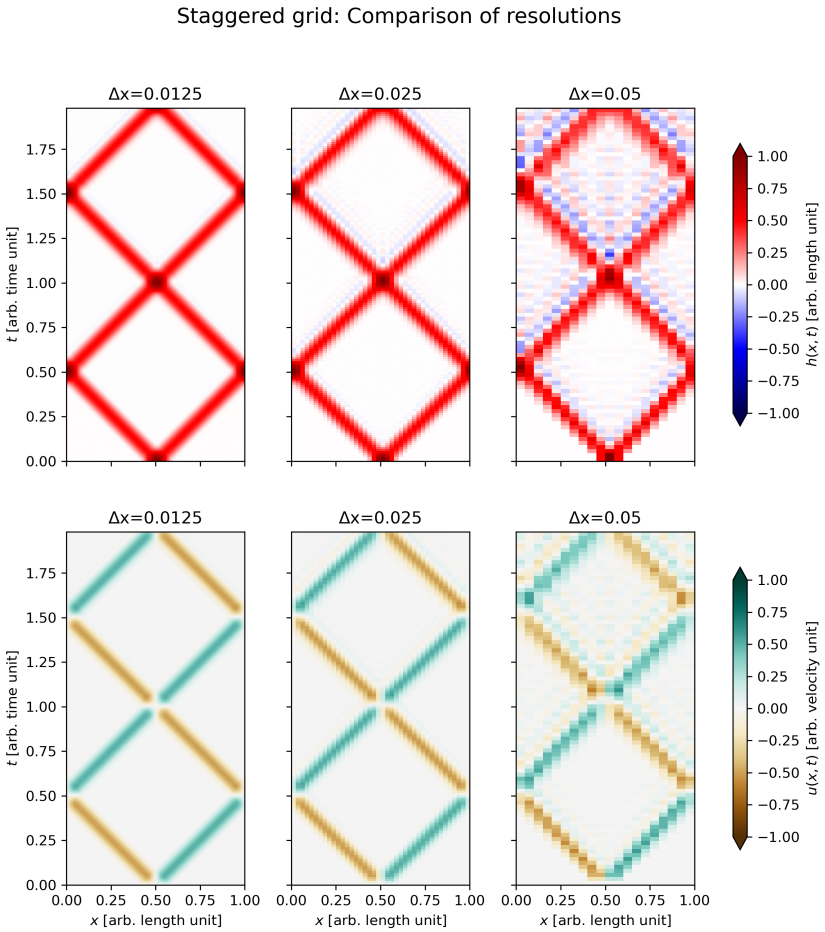

Figure 2: Comparison of staggered grid solutions at different spatial resolutions.

In Figure 3 one can see the time evolution of the relative error in h for the different numerical schemes. The error is shown for the unstaggered grid scheme and the staggered grid scheme with the different spatial resolutions, together with the Robert-Asselin filter applied to the spatial resolutions of $\Delta x = 0.025$. Since the error is very noisy, a smoothed version of the error is also shown through a moving average filter.

Interestingly, staggering the grid gave a much smaller error than the unstaggered grid with the Robert-Asselin filter applied. Furthermore, the error of the staggered grid scheme with the coarsest spatial resolution ($\Delta x = 0.05$) is in the same magnitude as the unstaggered grid scheme with the grid resolution of $\Delta x = 0.025$. This shows the significant improvement that staggering the grid gives. Furthermore, applying a Robert-Asselin filter to the staggered grid scheme with showed to lower the error even more, although the effect of the filter is not as significant as as lowering the grid resolution to the finest spatial resolution of $\Delta x = 0.0125$.

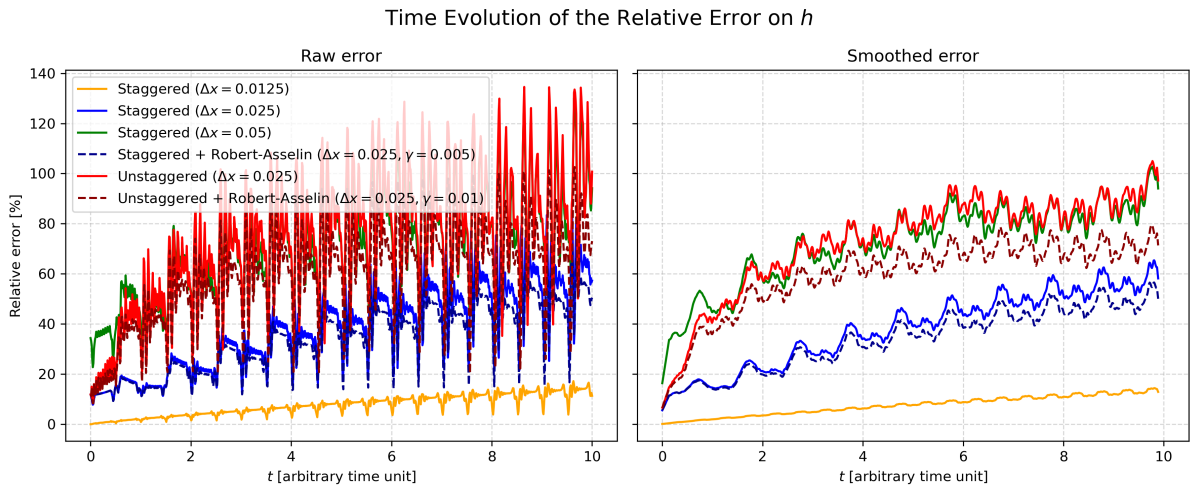

Figure 3: Time evolution of the relative error in h for different numerical schemes. The smoothed error curves uses a moving average filter and are shown for clearer visualization.

4 Discussion

The result presented above showcased that staggering the grid gave a much better agreement with the theoretical solution, compared to the unstaggered grid scheme. Interestingly one could see that the error of the staggered grid scheme with the coarsest spatial resolution ($\Delta x = 0.05$) is in the same magnitude as the unstaggered grid scheme with the grid resolution of $\Delta x = 0.025$. This result can be used from two points of view. First, this means that when computing a numerical scheme with a unstaggered grid, one can obtain the same error with a staggered grid scheme by using a twice as large grid spacing. This will thus save computational space and time, since the number of grid points will be halved. Another way of viewing this result is that by using the same grid spacing, staggering the grid will give a much smaller error than using an unstaggered grid. This can thus be used to improve the accuracy of numerical schemes.

One drawback of using a staggered grid scheme is that the scheme requires a smaller Courant number for stability, compared to the unstaggered grid scheme. This means that the time step needs to be smaller, which will increase the computational space and time. However, it was shown that the staggered grid could produce the same error as the unstaggered grid with a twice as large grid spacing, which means that although the scheme uses a smaller time step, it can still save space and time through the use of a larger grid spacing. Thus, although the staggered grid scheme is more limited by the Courant number, it can still be more efficient due to the lowering of the numerical error.

Regarding the Robert-Asselin filter, it showed to lower the error of of both the unstaggered and staggered grid schemes. However the affect of the Robert-Asselin on the unstaggered grid scheme did not lower the error as much as staggering did. However, the Robert-Asselin filter could also be applied to the staggered grid scheme, and showed to lower the error even more, although the effect was of a smaller scale. Thus, the Robert-Asselin filter can be used together with a staggered grid scheme to further lower the error of the numerical scheme.

5 Conclusion

In conclusion, the numerical solution of the shallow water equations was implemented using both an unstaggered grid and a staggered grid using the leapfrog scheme. The result showed that staggering the grid gave a much better agreement with the theoretical solution, compared to the unstaggered grid. It was shown that a twice as large grid spacing in the staggering gave the same accuracy as the unstaggered grid with normal grid spacing. Thus staggering the grid could be used to save computational space and time, while still maintaining the same accuracy or to improve the accuracy of the numerical scheme. The Robert-Asselin filter showed to lower the error of both the unstaggered and staggered grid schemes, although the affect on the unstaggered grid scheme was not as significant as staggering the grid. Thus, the role of using staggered grids in numerical schemes is an important tool to improve the accuracy of the model. This explains why staggered grids are more commonly used in weather models.

References

Kristofer Döös, Peter Lundberg, and Aitor Aldama Campino. *Basic Numerical Methods in Meteorology and Oceanography*. Stockholm University Press, Stockholm, Sweden, 2022. ISBN 978-91-7635-172-7. doi: 10.16993/bbs.

A Acknowledgements

The text in this report was spell-checked using Mistral AI's Le Chat. The schemes in this report were obtained using the Fortan programming language, while the figures were created using the Python programming language.